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Cultivation of anaerobes: from the basics to modern approaches

Cultivation of anaerobic bacteria: Foundations and principles

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ABSTRACT

A brief history of techniques in anaerobic microbiology are presented leading up to the incorporation of several improvements we have used over the years to improve our culture of anaerobic microorganisms of environmental, industrial and clinical importance. Two overriding aspects from our combined 90 years of experience here are: the better one's control of anaerobic conditions and gas phases, the better results are obtained; techniques can and should be targeted for individual microorganisms and accompanying experiments. Continued improvements in anaerobic microbiology are expected and encouraged for the future.

1. Introduction: anaerobic culture technique

Anaerobic microorganisms play a significant role in global cycling of elements and nutrients and play critical roles in health and disease processes in animals and humans [1–5]. In addition, many anaerobes are used in industrial and commercial applications [6–9]. Strictly anaerobic microorganisms are predominant members of a wide range of anoxic ecosystems such as freshwater and marine sediments and soils, and the gastrointestinal tracts of insects, birds, animals and humans. Indeed, it has been reported that 90 % of prokaryotic microbes reside in anaerobic environments and the majority remain to be isolated in culture [5,10]. Although Antonie van Leeuwenhoek noted that some "animalcules" were able to live in the absence of air, the greatest impediment to the investigation of anaerobes were laboratory methods to create an oxygen-free atmosphere. Hence, it was not until 1860 and Pasteur's investigations into wine spoilage that the first techniques of anoxic cultivation methods and medium preparation were developed. It is pertinent to note that he used the terms 'aerobic' and 'anaerobic' for the first time [11,12] (see Fig. 5).

As the basic oxygen-free culture methods of Pasteur were improved upon, the inclusion of agar into medium in the 1880s for surface culture enabling single-colony isolation proved to be key to the cultivation of anaerobes [13]. A pivotal step forward in anaerobic culture technique was developed by the pioneering work of Robert E. Hungate [14]. His study of cellulose-degrading microorganisms in the bovine rumen in 1944 led to the revolutionary roll-tube approach that enabled the successful culture of *Clostridium cellobioparus*. These methods have been used in many laboratories globally with additional modifications and have been covered extensively in the literature [14–19]. Hungate technique expanded the isolation and characterization of anaerobes, principally using a cannula to flush N_2 gas into the culture vessel to remove oxygen before sealing the vessel; this basic Hungate technique is still used today (Table 1 [17,19]). However, the method itself was limited in being time-consuming and laborious, and improvements were adopted by workers in the field. Some aspects of the Hungate technique, such as the original Hungate tube (Fig. 1a) and Hungate tube press [17, 19] [Table 1], and the use of alkaline pyrogallol in a Bray dish (Fig. 1b) have been essentially abandoned, while the venerable candle jar or use of plates of O_2-consuming bacteria to establish anoxic conditions remain relevant [20]. Modifications of the Hungate technique developed by Holdeman, Cato and Moore [16] included growth of anaerobes with N_2 and CO_2 gas phases. Other technical tips such as the preparation of hemin and vitamin K stock solutions are also given here [21]. Miller and Wolin [22] used serum bottles closed with a butyl rubber stopper with a crimped metal seal, a method still favored by many modern-day laboratories.

A major contribution was made in the mid 1970s by William E. Balch and Ralph S. Wolfe, who utilized the use of pressurized tubes (later referred to as Balch tubes and Balch technique) in combination with gassing stations and culture transfer using anoxic sterile syringes [23–25]. These techniques employ user-friendly glassware and equipment to easily manipulate anaerobes or microorganisms requiring a defined gas phase. This method also allowed the ability to handle 200–300 cultures daily. Alumni of the Wolfe laboratory helped

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Table 1
General isolation and cultivation references.

Hungate technique	anaerobic technique based on manipulation of materials under sterile, anoxic gas; roll tube isolation	R.E. Hungate, A roll tube method for cultivation of strict anaerobes, Methods Microbiol. 3B (1969) 117–132.
Hungate technique	thorough overview of Hungate technique	M.P. Bryant, Commentary of the Hungate technique for culture of anaerobic bacteria, Am. J. Clin. Nutrition (1972) 1324–1328.
Balch technique	use of pressurized gas phase; culture transfer using anoxic sterile syringes	W.E. Balch, R.S. Wolfe, New approach to the cultivation of methanogenic bacteria: 2-mercap-toethanesulfonic acid (HS-CoM)-dependent growth of *Methanobacterium ruminantium* in a pressurized atmosphere, Appl. Environ. Microbiol. 32 (1976) 781–791.
Balch technique	further development of Balch technique, including media, plate incubation and larger-scale culture	W.E. Balch, L.J. Magrum, G.E. Fox, R.S. Wolfe, C.R. Woese, Methanogens: Reevaluation of a unique biological group, Microbiol. Rev. 43 (1979) 260–296.
Anerobic technique	general technique focused on clinical anaerobes	H.R. Jousimies-Somer, P. Summanen, D.M. Citron, E.J. Baron, H.M. Wexler, S.M. Finegold, Wadsworth-KTL Anaerobic Bacteriology Manual, 6th ed., Star Publishing Co., Belmont, CA, 2002.
General technique	general cultivation with some emphasis on anaerobes	R.S. Tanner, Cultivation of bacteria and fungi, in: C.J. Hurst, R.L. Crawford, A.L. Mills, J.L. Garland, L.D. Stetzenbach, D.A. Lipson (Eds.), Manual of Environmental Microbiology, 3rd ed., ASM Press, Washington D.C., 2007, pp. 62–70.
Uncultured ruminal bacteria	includes targeted, novel medium	N. Kenters, G. Henderson, J. Jeyanathan, S. Kittelmann, P.H. Janssen, Isolation of previously uncultured rumen bacteria by dilution to extinction using a new liquid culture medium, J. Microbiol. Methods 84 (2011) 52–60.
Anaerobic technique	isolation of uncultured microorganisms; compact generators for controlled atmospheres	T. Narihiro, Y. Kamagata, Anaerobic cultivation, in: M.V. Yates, C.H. Nakatsu, R.V. Miller, S. D. Pillai (Eds.), Manual of Environmental Microbiology, 4th ed., ASM Press, Washington, D.C., 2016, pp. 2.1-2.1 – 2.1-2.12.
Anaerobic technique	review focused on strict anaerobes	N. Hanisakova, M. Vitezova, S.K.-M.R. Rittmann, The historical development of cultivation techniques for methanogens and other strict anaerobes and their application in modern microbiology, Microorganisms 10 (2022) 412–439.

popularize this method and it became the method of choice of many laboratories (including the Department of Microbiology and Plant Biology at the University of Oklahoma, recognized as a leader in anaerobic microbiology from the 1980s).

Initial studies with the rumen ecosystem and the isolation of anaerobes provided an impetus to study anaerobes associated with humans. The contributions of Holdeman, Moore, Cato and colleagues at the Anaerobe Laboratory, Virginia Polytechnic Institute and State University in the 1970s cannot be overstated. It is especially important to mention that this group not only described an incredible number of novel anaerobes from human sources [26–31] but provided the first insights into the human microflora (or the microbiome, as we now call it) [32]. These studies encouraged clinicians and clinical microbiologists to investigate human pathogens building upon the work of Pasteur and Joubert in 1877 [12,33] who described the first anaerobic pathogen *Clostridium septicum*. However, in the mid 20th century there was a general lack of interest in anaerobes and their role in infections (for a review see Finegold,1993 [34]. It was not until the 1950s that the pioneering work of Sydney Finegold began to reveal to clinicians to the importance of anaerobes in clinical infections [35,36]. Finegold and co-workers described over many decades a plethora of novel anaerobic taxa recovered from clinical material [34,35,37–40]) and Finegold also provided some of the first data on the effect of diet on the human fecal microbiota [37]; he was also instrumental in the important reclassification of the important human pathogen *Clostridium difficile* to *Clostridioides difficile* in 2016 [38]. Furthermore, he was co-author of the Wadsworth-KTL Anaerobic Bacteriology Manual published in 2002 and still an invaluable text for any laboratory involved in anaerobic microbiology [39]. Table 1 also provides some other key references in anaerobic microbiology.

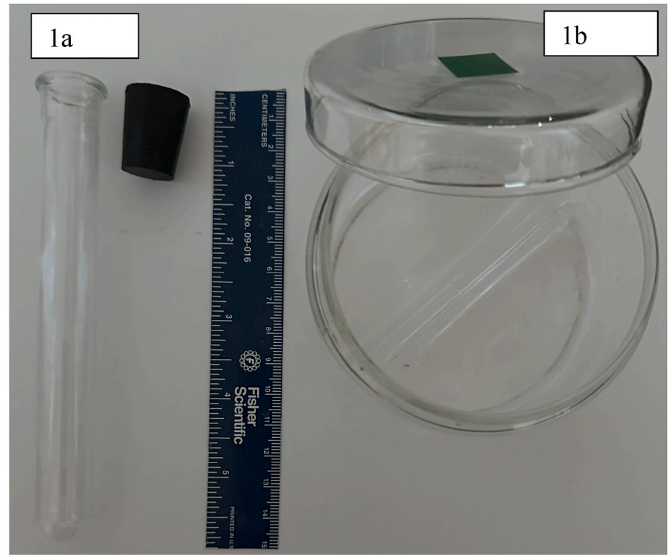

Fig. 1A. Hungate tube; 1b. Bray dish.

1.1. Anoxic environment in the laboratory – practical considerations from our experience

The Balch technique, which uses tubes and serum bottles combined with anaerobic chambers and jars, are the primary methods of choice used in the Lawson (animal and human microbiomes) and Tanner laboratories (biofuels, biocorrosion, environmental microbiology); our combined efforts have resulted in the description of a wide range of anaerobic bacteria, including new genera [40–47]. The procedures developed by Balch and Wolfe will be the focus of the discussion here. The work by one of us (RST) describing the genus *Acetobacterium* [48] is of interest as this was when he transitioned from the Hungate to the Balch technique. Again, some of the key methodological references for anaerobic microbiology are in Table 1.

Simple approaches of anaerobic technique [Jousimies-Somer et al., 2002] (Table 1) can be used successfully for many clinically important anaerobes, which, in general, are less sensitive to O_2. For example, the oxygen tolerance of clostridial species ranges from relative insensitivity, such as for *Clostridium aerotolerans* [49] to extreme sensitivity, such as for the hydrogen-oxidizing acetogens [50]. However, the better one can control anoxic and redox conditions, the more control one has over experimental outcomes. *Clostridium carboxidivorans* [41] can grow in the presence of 1 % O_2, but ppm levels of O_2 completely eliminate a desired phenotype such as ethanol production (data not shown). *C. carboxidivorans* can also be routinely cultured in a defined medium in which a stock solution containing PABA, biotin and pantothenate can replace yeast extract and a vitamin solution [Tanner, 2007, Table 1] often used to grow this acetogen (data not shown). Similarly, culture of *Haloanaerobium* spp. does not require a reducing agent (data not shown). This would not be possible without the use of strict anaerobic technique.

A good grade of compressed gas, such as prepurified grade or better, is the starting point for provision of anoxic atmospheres. Gas can be scrubbed free of O_2 in a heavy-walled copper tube (2.5 x 20–30 cm) packed with copper turnings and heated to 150–200 °C in a tube furnace or with heating tape. Periodic exposure of this scrubber to H_2 will readily regenerate the system. Our Hungate probes (Fig. 2) are constructed from 2 cc glass syringe barrels with luer lock tips and packed with nonabsorbent cotton. Stainless steel needles, 17 gauge (Z101125, Sigma-Aldrich, St. Louis, MO), complete the probe. These tolerate many flame sterilizations, are small enough to be used with stoppered glassware without chipping it but have a large enough bore for flushing syringes prior to anoxic transfers.

A good anaerobic chamber equipped with an air lock greatly facilitates preparation of materials for anaerobic microbiology. We have found the flexible chambers from Coy Laboratory Products, Inc. (Grass Lake, MI) the easiest chambers to use and maintain [Tanner, 2007, Table 1], although solid framed chambers from Don Whitley Scientific (UK) and Anaerobe Systems (CA) are also popular. The advantage of the anaerobic chamber is that medium and anaerobes may be handled just like on an open bench, facilitating plate transfers, dispensing of medium etc. Our chambers are operated with a gas phase of 1–2 % H_2 in N_2; O_2 levels of 2 ppm or lower are (as measured by a palladium hydrogen probe) are easily achieved. Depending on the taxa being investigated (gut anaerobes, microaerophiles (host derived and environmental) some laboratories can use alternative gas combinations including an increased percentage of CO_2, however the use of CO_2 can bring its own challenges as outlined below in section General Medium Preparation – specific requirements for anaerobes.

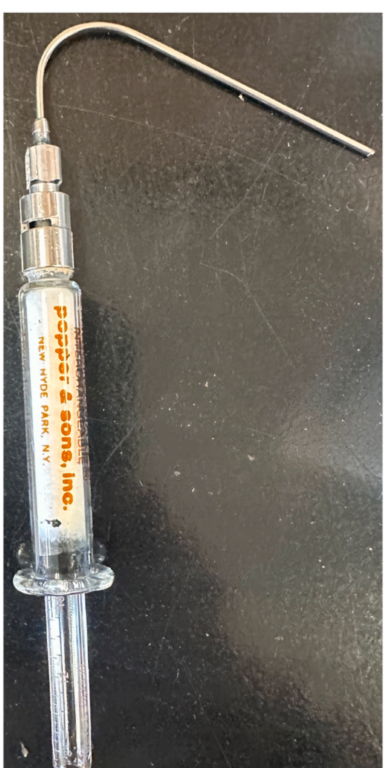

Fig. 2. Hungate probe.

Inclusion of activated charcoal gas scrubbers and an agent to remove moisture from the atmosphere (anhydrous calcium chloride pellets, 4–20 mesh) facilitate chamber and chamber catalyst (palladium on alumina pellets; 205745, Sigma-Aldrich) maintenance. An anoxic solution of 1 N HCl is used to clean and disinfect the anaerobic chamber. This leaves no residue, and its vapors are absorbed by the calcium chloride drying agent. Chambers seem to be easier to maintain if they are not used for culture incubations as the primary byproducts of anaerobic bacterial growth are organic acids like lactic, succinic, propionic, butyric and acetic acids, and sometimes ethanol. Such products are not only corrosive to the chamber itself but also cause damage to equipment (fans, metal forceps etc). Furthermore, if not controlled these products can lead to the growth of mold and other contaminants on the chamber walls). The PVC flexible bags of our chambers last more than 10 years (often much longer), even with daily use.

A gas exchange station or gassing station (Fig. 3) [Balch and Wolfe, 1976 [24], Table 1] allows control over the final atmosphere used for tube or plate culture of anaerobes. Elimination of as many joints and valves as possible, as well as construction from stainless steel components, eases the use and maintenance of a gas exchange station. Many of our gassing stations were installed in the early 1980s but have been modified over time. We are not averse to taking ideas from colleagues to improve the efficiency of our systems. For example, in Figs 3, 4 and 5; we have adopted an improved manifold fabrication that reduces many connections, sources of leaks of the gassing station combined with a more robust terminal needle configuration.

1.2. Growth medium and anaerobiosis

Removal of O_2 and lowering the redox potential of a medium by the addition of a reducing agent are two crucial parts of anaerobic microbiology. The removal of O_2 from a medium is usually achieved by boiling the medium (this eliminates any dissolved gas). This will often be performed with the medium under a stream of anoxic gas [19] for a small batch of medium, or by steam sterilization in an autoclave for a large batch of medium. Steam sterilization is performed with a vessel which is covered but still open to the atmosphere. After autoclaving, the medium is cooled under a stream of sterile, anoxic gas with gentle swirling, then sealed [23]. Finally, sterile reducing agent is added to reduce the medium's redox potential. A variety of reducing agents are available, many based on sulfur chemistry [51–53]. Sodium sulfide solutions are commonly used. We use a cysteine·sulfide solution for most of our routine culture work. Cysteine (2 g) (free base cysteine, not cysteine hydrochloride or another acid salt) and anoxic water (boiled under an oxygen-free gas, then stoppered) are placed in an anaerobic chamber overnight. Sodium sulfide nonahydrate crystals (2 g, freshly rinsed and

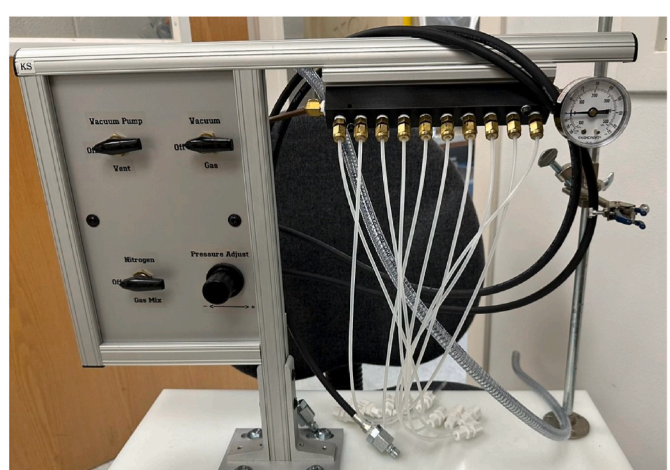

Fig. 3. Gas exchange station (Gassing Station).

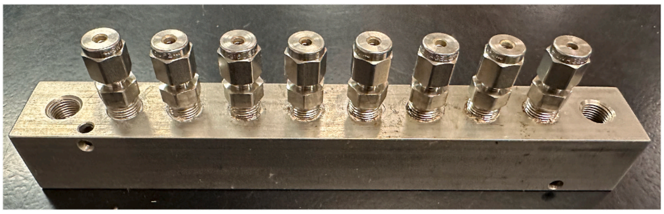

Fig. 4. Right-angle flow rectangular manifolds (McMaster-Carr, Elmhurst, IL). Learned from a colleague at OU.

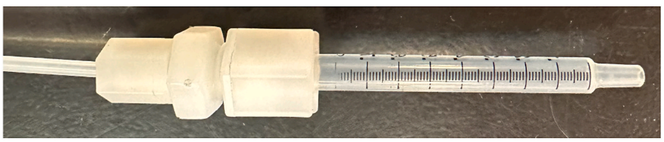

Fig. 5. 1 cc syringe barrel terminal hardware. Learned from a colleague at the University of Georgia. The tubing is 1/8″ multilayer tubing (cat. no. T4203, Qosina, Ronkonkoma, NY).
The syringe is attached to a needle which can be inserted through a rubber septum in Balch tubes or serum bottles.

dried) are taken into the chamber and dissolved with the cysteine in 100 mL of water. This is dispensed and stored in sealed tubes. For most media, 0.1–0.5 mL of this reducing agent is added per 100 mL of degassed medium taken into an anaerobic chamber for dispensing prior to sterilization. An advantage of this reducing agent is that it is stable when prepared thusly for over four years, if stored in an anaerobic chamber.

1.3. General Medium Preparation – specific requirements for anaerobes

In addition to the anoxic environments, an important consideration is the use of appropriate media to provide the nutrition used to recreate the natural habitat and source material of the samples being investigated. In addition to nutrients (carbohydrates, proteins, blood, serum), many organisms require metals and/or vitamins. Other factors to consider are incubation temperature and time, pH, etc.

A general base culture medium and the composition of stock solutions used to prepare it are given in Tables 2–5 (Tanner, 2007, Table 1). There are several other formulations of mineral, vitamin and metal stock solutions, many of which may be found in Atlas (2019) [54]. The desired carbon/energy source and electron acceptor, if any, are added to this base medium for enrichment, isolation and culture of desired cell types. Attention to the type of microbial metabolism desired will indicate modification of this or any other basal medium recipe. For example, if a nitrogen-fixer were desired, ammonium salts would be eliminated from the medium. If strict control of any potential sulfate-reducing bacteria were required, chloride salts would substitute for the sulfate salts in the recipes.

Table 2
Base for defined/semi-defined anaerobic media (/L)[a].

Mineral solution	10 ml
Vitamin solution	10 ml
Trace metal solution	5–10 ml
Buffer[b]	5–20 g
adjust pH with NaOH	

[a] Tables 2–5 adapted from Tanner 2007 (Table 1). Yeast extract (0.1–2 g/L) useful in many media.
[b] TES (pK_a 7.4) for neutral pH, MES (pK_a 6.1) for slightly acidic pH, TAPS (pK_a 8.4) for slightly alkaline pH, $NaHCO_3$ (added after pH adjustment) for media with CO_2 in the gas phase.

Table 3
Mineral solution for anaerobic media (g/L)[a].

NaCl	80
NH_4Cl	100
KCl	10
KH_2PO_4	10
$MgSO_4 \cdot 7H_2O$	20
$CaCl_2 \cdot 2H_2O$	4

[a] Analytical reagent grade chemicals recommended. Add and dissolve in order. Note the relatively high amount of ammonium and low amount of phosphate. Some microorganisms require more phosphate, such as lactic acid bacteria and thiobacilli, which can also function as a buffer. Stable at room temperature.

Table 4
Vitamin solution for anaerobic media (mg/L)[a].

Pyridoxine · HCl	10
Thiamine · HCl	5
Riboflavin	5
Calcium pantothenate	5
Thioctic acid	5
p-Aminobenzoic acid	5
Nicotinic acid	5
Vitamin B_{12}	5
MESA	5
Biotin	2
Folic acid	2

[a] Note the relatively high amount of vitamin B_{12}, and the presence of MESA (mercaptoethanesulfonic acid), required by some methanogens. Store under refrigeration.

Table 5
Trace metal solution for anaerobic media (g/L)[a].

Nitrilotriacetic acid[b]	2.0
$MnSO_4 \cdot H_2O$	1.0
$Fe(NH_4)_2(SO_4)_2 \cdot 6H_2O$	0.8
$CoCl_2 \cdot 6H_2O$	0.2
$ZnSO_4 \cdot 7H_2O$	0.2
$CuCl_2 \cdot 2H_2O$	0.02
$NiCl_2 \cdot 6H_2O$	0.02
$Na_2MO_4 \cdot 2H_2O$	0.02
Na_2SeO_4	0.02
Na_2WO_4	0.02

[a] Note the presence of selenium and tungsten, and the absence of boron (required by cyanobacteria) and aluminum. Store under refrigeration.
[b] Adjust pH to 6.0; this adjustment requires some time to achieve.

The growth of many anaerobes, even those which produce significant amounts as an end-product, is stimulated by the presence of CO_2 in the gas phase. This CO_2 must be balanced for the desired culture pH by bicarbonate or carbonate. In general, the higher the pH, the higher the incubation temperature and the higher the partial pressure of CO_2 in the gas phase, the more bicarbonate required. For example, if using a gas phase containing 20 % CO_2 at a gauge pressure of 200 kPa for culture at pH 7.0, we will add 2 g/L $NaHCO_3$ for incubation at 37 °C or 5 g/L for incubation at 60 °C. For cultivation at pHs of 6.0 or lower, we omit bicarbonate addition (one is outside of the bicarbonate/CO_2 buffer range). An advantage of using gas phases which contain 10–30 % CO_2 is that the amount of bicarbonate which needs to be added to the medium is small enough so that it can be sterilized as part of the whole medium and doesn't need to be added from a sterile, anoxic stock after sterilization. If an 100 % CO_2 atmosphere is used, it is often necessary to add bicarbonate/carbonate from a sterile stock solution to a medium after steam sterilization to avoid precipitation or other problems.

The early studies of Marvin P. Bryant and co-workers investigating the rumen provided advances in media preparation (Bryant, 1972, Table 1) [19,55–57]. Of particular note is the addition of clarified rumen fluid as a nutritional supplement, still routinely used in the cultivation of anaerobes recovered from animal and human gastrointestinal tracts via fecal material. The composition of bovine ruminal fluid is characterized by phospholipids, inorganic ions, gases, amino acids, dicarboxylic acids, fatty acids, volatile fatty acids, glycerides, carbohydrates and cholesterol esters; many are microbial fermentation products that occur under the anaerobic condition found in the rumen [58,59].

The basal medium described in Table 2 can be adapted using different substrates and electron acceptors for the culture of many different microorganisms [60]. Of course, if an excellent medium for this purpose is already published it should be used, such as for iron-reducing bacteria [61].

Another advantage of our approach to medium preparation is that all the medium components can be added at the beginning of preparation, followed by degassing, reduction (if necessary), dispensing, sealing, gas exchanging and autoclaving, without the need to add separately sterilized components later. Although preparation of a medium in this way may lead to precipitation of certain substrates from the medium, this is temporary and will resolve. For example, *Methanothermobacter* medium contains a high amount of $NaHCO_3$ (5 g/L), added after pH adjustment of the medium to balance CO_2 in the gas phase. After autoclaving, a precipitant will be in this medium due to loss of dissolved CO_2 caused by the heating. However, as the medium cools and CO_2 goes back into solution, the precipitate dissolves, clearing the medium. Another advantage of preparing anaerobic medium as described in Tanner, 2007 (Table 1) is stability of the cysteine · sulfide reducing agent compared to a sulfide solution alone. A sign of sulfide solution instability upon storage is the appearance of a precipitate.

Use of semi-defined medium can result in greatly improved culture/enumeration of various microorganisms. Examples include a methanogen [62]), sulfate-reducing bacteria [63,64]) and acid-producing bacteria [65]). Medium optimization may also be performed for improving a process, such as alcohol production [66].

Our method of choice, the Balch technique may not always be the best for growth of some anaerobes. Pure culture growth of syntrophs using the Hungate technique, flushing for all manipulations, is recommended, as even gas exchange may not remove enough H_2, present in an anaerobic chamber (ppm levels of H_2 can inhibit their growth) [67].

1.4. Long-term storage of anaerobes

There are several ways to prepare cultures of anaerobes for long-term storage. While lyophilization is recommended, cryopreservation techniques using commercial kits or liquid nitrogen are also useful [68]. A relatively easy method that has been successful in our hands is stocking in glycerol [69]. Anoxic glycerol (2 mL) is placed in serum-seal vials (10 mL) under a N_2 gas phase and sterilized. A cell pellet is collected after gentle centrifugation (this can be performed directly in serum-seal tubes with appropriate adapters). The cell pellet is resuspended in 2 mL of either fresh or spent medium and injected into the vial. Vials are stored at −20 °C; these will stay liquid, so there will be little freeze damage to the cells. One can readily concentrate cells from as much as 20 mL of broth culture for this 2 mL addition.

There are many sources of information on handling specific groups of anaerobes. One example is for clostridia [70,71]. Even here, awareness of the literature beyond these sources is important. For example, when dealing with toxigenic strains of clostridia, including sugar such as sucrose or glucose in a medium can inhibit toxin production, making working with these strains safer [72,73].

Finally, if one is not experienced in the culture of anaerobes, visiting a lab that routinely does this is a good investment, especially one working with those microorganisms that you are interested in. The authors have decades of experience training microbiologists in the academic, private and government sectors on how to handle anaerobes.

1.5. New developments in cultivation of anaerobes

The application of modern sequencing methods combined with culture-independent approaches has been a double-edged sword for the isolation and description of novel microorganisms. These methods have undoubtedly provided unprecedented insights into the diversity, structure and understanding of microbial communities across a broad range of ecosystems [2,4,74–81]. Sequence-based approaches and the associated bioinformatic programs now predominate with many taxonomies relying to an increasing extent on genome-based phylogenies [82–87]. However, such approaches have resulted in unintentional consequences with respect to cultivation of microorganisms. Many principal investigators now prefer culture-independent, high throughput molecular methods to describe novel taxa, and in many instances, using minimal phenotypic characterization, dispensing with cultivation completely [88]. This is especially true for the cultivation of anaerobes due to the increased time demanded, specialized equipment and technical difficulty of culturing these organisms. Indeed, many organisms described as novel taxa have no cultured representative, being identified only by a genome sequence; such organisms are given the status “Candidatus” (as introduced to bacterial taxonomy in the 1990s) [89–91]. Furthermore, many journals now discourage the publication of single-strain taxa; however, many species and genera have been described on a single isolate, many were published in the 1990s and early 2000s as single strain taxa and many remain as single strain taxa (e.g. *Marvinbryantella* [92], *Fontimonas* [93], *Ruminococcus luti* [94], *Tolumonas osonensis* [40], *Cloacibacterium* [95], *Alkalibaculum* [96]. If such organisms were removed from List of Prokaryotic names with Standing in Nomenclature LPSN (LPSN.dsmz.de) a huge amount of bacterial biodiversity would be lost!

A recent renaissance and interest in cultivation *per se* that includes anaerobes has led to further developments and refinement of anaerobic methods. In part, this renewed interest is due to the huge interest in the human microbiome and its association with human health and disease [97]. The majority of the gut microbiome is composed of mainly anaerobic organisms, with several groups having made important contributions to the identification and distribution of anaerobic taxa [74,75,98]. Gordon and colleagues have made a tremendous impact on our understanding of the human gut microbiome using anaerobic culture initiatives combined with 16S rRNA sequencing and gnotobiotic mouse models. Mice colonized with a complete human microbiota demonstrated major changes in microbial community structure when subjected to a change in diet from low-fat, plant polysaccharide-rich to a high-fat, high-sugar Western diet [99]. Another major advance is what was to become known as culturomics, where initially a wide range of selective culture conditions are used [100] followed by a polyphasic approach; bacterial isolates are first identified using MALDI-TOF MS, followed by 16S rRNA gene and whole genome sequencing which is compared to international databases. Phenotypic data is then incorporated to produce a genus/species description. This approach has been successful in the recovery and description of many novel anaerobic organisms [101–104]. However, it must be noted that many taxa described in the literature have not been validly published and thus have no standing in the literature.

1.6. Old meets new

Despite our best efforts, many taxa remain recalcitrant to cultivation as demonstrated by molecular techniques [105]. One approach to investigate the diversity of microbial communities is the sequencing of genomes using culture-independent single-cell genomics [106–109]. Similarly, such approaches are now being coupled towards the culturing of anaerobes. Somewhat ironically, this state-of-the-art approach is rooted in the classical method of most probable number (MPN) and dilution to extinction methods [110,111] that have been successfully used in the isolation of novel anaerobes for decades (Kenters et al., 2011, Table 1) [95,112–114]. For high-throughput methods, one approach has been to estimate the total cell numbers by microscopy or flow cytometry, the samples are then diluted in tubes or 96/384-well plates so on average 1 cell per tube/well. This method has been applied to the isolation of anaerobes from the rumen and the human oral cavity [112,115]. More recently, Clavel and co-workers described an elegant method for an anaerobic single-cell dispensing method that greatly aided the cultivation of anaerobes from the human gut [116].

2. Conclusions

In closing, the isolation of organisms in pure culture permits the elucidation of the physiological and metabolic capabilities of different microorganisms. This is particularly important in ecological and microbiome studies especially when interpreting the gene content of genomes; gene expression is greatly influenced by environmental conditions, thus determining the phenotype observed. A gene may be present in the genome but never expressed, or more significantly it's when the gene is expressed in a complex ecosystem that provides a better insight into microbe-microbe and microbe-host interactions [117–119]. Therefore, we are strong proponents of cultivation studies, augmented by genomic analysis and the biology they reveal.

CRediT authorship contribution statement

Paul A. Lawson: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Ralph S. Tanner:** Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare we have no financial and personal relationships with other people or organizations that could inappropriately influence (bias) our work.

Data availability

The data that has been used is confidential.

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